THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING: LEVEL 1 EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

ENGINEERING MECHANICS AND MATERIALS

- CIVL1013 (3-year Curriculum)
- CIVL1113 (4-year Curriculum)

DATE: 27, December 2012 TIME: 2:30 pm – 5:30 pm (3 hours)

Answer any 2 questions from Section A and any 2 questions from Section B.
All questions carry equal marks.
Use a separate answer book for Section A and Section B.

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

SECTION A

You are required to answer 2 out of 3 questions in Section A.

Question 1

(a) Derive the following relationship from first principals. You may use diagrams to assist in your derivation. Also, what is the physical interpretation of κ? **(10 marks)**

$$\kappa = \frac{1}{\rho} = \frac{d\theta}{dx}$$

(b) Considering the beam in Figure 1, (i) calculate the reactions (with suitable checks), (ii) draw the shear force and bending moment diagrams, and (iii) calculate the maximum shear force and bending moment and their respective locations. Note that the tension side bending moment diagram is required. **(15 marks)**

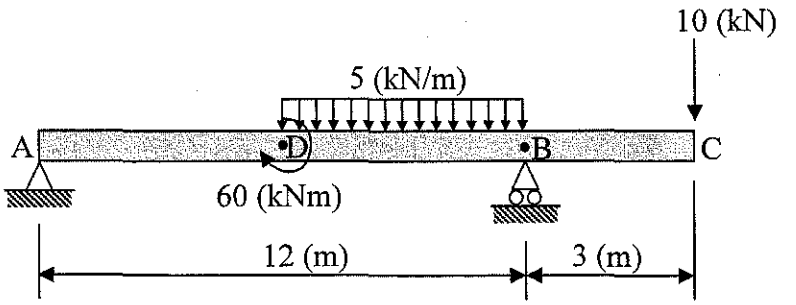

Figure 1. (note: Point D is mid-span of A-B)

Question 2

(a) Derive the following differential equation from first principals. You may use diagrams to assist in your derivation. **(10 marks)**

$$\frac{d^2v}{dx^2} = \frac{M}{EI}$$

(b) Calculate the reaction of support B for the propped cantilever beam shown in Figure 2 (hint: the vertical movement at support B is zero). **(15 marks)**

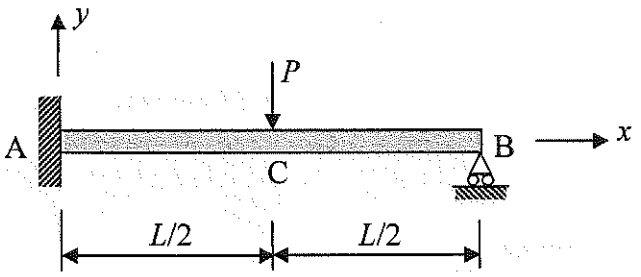

Figure 2.

Question 3

(a) Derive the equation of Mohr's circle from the transformation equations. **(10 marks)**

(b) Construct Mohr's circle for the element shown in Figure 3. Make sure to include all necessary information on the Mohr's circle diagram. Then, plot the state of stresses on the same diagram for an element inclined in an anti-clockwise direction of θ degrees. **(15 marks)**

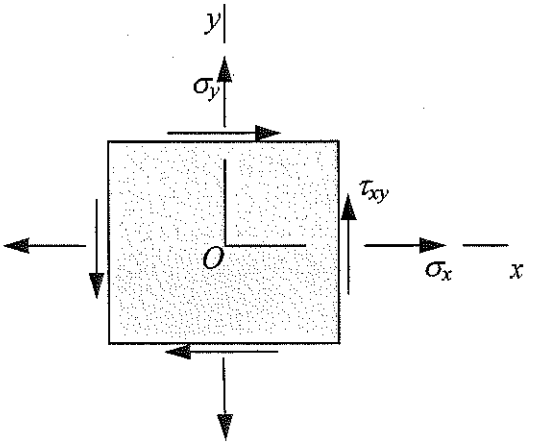

Figure 3.

- END OF SECTION A -